

Silk Road Journey: A Real-time AI-based Interactive Art Installation for Silk Road Cultural Reenactment and Experience

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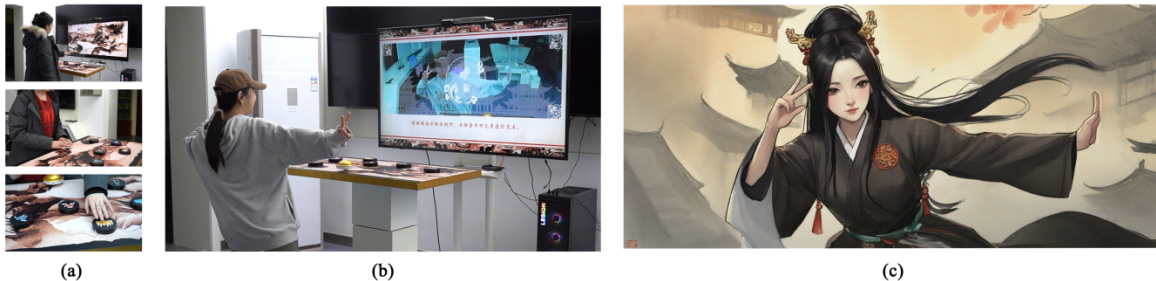


Figure 1: The overview of Silk Road Journey we presented. (a) Demonstration and motion interaction. (b) Real-time physical interaction with countdown screen. (c) AI generated stylized image of Chang'an

ABSTRACT

The Silk Road, spanning thousands of years, possesses splendid civilization relics. However, with the disappearance of historical records, there is a certain lack of public awareness of the Silk Road culture. Silk Road Journey is an AI interactive installation based on six significant cities along the Silk Road, allowing the audiences travel through different eras, and pose with the scenes through physical interaction to generate unique commemorative images. We apply AIGC technology to the design of the interactive art installation related to cultural heritage. We employ a variety of strategies to improve the efficiency and quality of gesture-controlled image generation, aiming to improve the immersion of the interaction and achieve real-time interaction effects. This innovative way of combining new technology, interactive experience and world cultural heritage improves the public's cultural awareness of the Silk Road,

which also seeks out a new mode of experience and dissemination for cultural heritage preservation.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Human computer interaction (HCI); Interaction techniques; Gestural input; • **Computing methodologies** → Artificial intelligence.

KEYWORDS

Silk Road Cultural Heritage, AIGC(Artificial Intelligence Generated Content), gesture recognition, embodied interaction, diffusion model

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1 INTRODUCTION

The Silk Road, as a world cultural heritage, was an ancient trade network connecting countries in Europe, Asia, and Africa. However, a significant amount of historical records related to the Silk Road have been lost or fragmented. Additionally, the preservation of Silk Road cultural heritage inherently involves cross-cultural communication. These factors have contributed to a certain lack of public understanding regarding Silk Road culture.

The emergence of AIGC (Generative Artificial Intelligence) has introduced new possibilities for cultural heritage presentation and dissemination, especially the diffusion models such as DALL-E and Stable Diffusion. AIGC technology provides a new way of artistic expression for promoting Silk Road culture, but also brings technical challenges. Our core work is to explore these possibilities. We have utilized AIGC to construct the Silk Road Journey interactive art installation. In this installation, audiences assume the role of travelers and engage in real-time interaction through physical gestures.

In summary, our research explores the following aspects: (1) Interactive artworks with the theme of the Silk Road: researching the literature and canonical texts related to Silk Road culture, exploring the artistic styles, legends and stories of the countries along the Silk Road, in order to promote the preservation of Silk Road cultural heritage. (2) Utilizing AI diffusion models to generate artistic materials for the installation; achieving real-time interactive effects by capturing photos based on the audience's poses and transforming them into illustrations with regional cultural characteristics. (3) Designing an interactive experience space that allows the audience to walk through the historical Silk Road, evoking a sense of association and participation, and at the same time establishing a deep recognition of the cross-cultural integration of Silk Road heritage.

2 RELATED WORK

2.1 AIGC and Cultural Heritage Preservation

After more than 60 years of evolution, artificial intelligence has entered a new stage of development. Artificial Intelligence Generated Content (hereinafter referred to as AIGC) has rapidly emerged as a significant application of AI. In addition to using textual prompts as a basis, diffusion models allow for diverse images as reference inputs, such as hand-drawn sketches, line drawings, color blocks, and pose expressions.

AIGC technology has also made positive contributions to the preservation and inheritance of cultural heritage. The powerful generative capabilities of artificial intelligence can support the restoration and reproduction of traditional art forms, simulating and preserving cultural heritage content [6]. For example, AIGC technology can be used to design reference images for Zisha teapots using MidJourney [8]. AIGC technology has also been applied to simulate and derive designs from traditional decorative patterns [12].

2.2 The Silk Road and Digital Preservation

In recent years, with the inclusion of the Silk Road in the World Heritage List, academic research related to the Silk Road has garnered attention and importance from scholars along its route.

In the context of the "Belt and Road Initiative" and the protection of world cultural heritage, the digital protection of the Silk Road is imperative. Taking the digital preservation of Dunhuang murals as an example, efforts have been made to establish "digital archives" for ancient architecture and cultural artifacts, exploring areas such as virtual restoration and online exhibitions.

2.3 Test Demo: Portal

To validate the feasibility of real-time AI generation, we created a four-panel interactive cartoon called "Portal", in which each panel's content is randomly generated based on the audience's actions. On the technical side, we used a computer camera to capture the image input and generated the corresponding comic image through Stable Diffusion. In terms of interactive experience, the audience is no longer passive observers but active participants, sharing the ups and downs of the story and enjoying the fun alongside the characters.

The demo provides reliable suggestions and references for the production of interactive installations. In the Silk Road Journey, we focus on enhancing cultural identity through the infusion of cultural themes, improving immersion with experiential spaces and physical interactive devices, and increasing the efficiency and quality of AI generation for faster image production.

3 SILK ROAD TRAVEL INTERACTIVE AI INSTALLATION

3.1 Theme and Cultural Connotation

The work selects representative cities along the land and maritime Silk Roads as key nodes. By searching literature and exploring local cultures and legendary stories, a complete picture book narrative is crafted for each location. This includes 4-6 pages of story plot illustrations and interactive elements that engage with the story. The selected representative locations include:

Chang'an and Dunhuang, which are important cities on the Silk Road. The legend of Xuanzang unfolds the Silk Road's scroll utilizing traditional Chinese painting techniques, while Dunhuang frescoes, especially the Legend of the Deer King, guide the generation of the illustrations. Persia and Istanbul thrived as artistic centers on the Silk Road. Persia's story draws from the literature of the "King of Birds" through decorative design, while Istanbul showcases bustling merchants with watercolor tracing. Cairo and Venice, important cities along the maritime Silk Road, offer interactive experiences. Viewers reunite with the Egyptian cat goddess in Cairo and glide through Venice's waterways in Egyptian mural and realistic oil paintings.

3.2 Interactive Design Strategies

In the Silk Road Journey interactive art installation, we aim to immerse viewers in the story space, as if they are traveling through time and distance, back to the historical Silk Road. Therefore, we set up an interactive sequence:

At the entrance of the installation, there is a long scroll that uses linear space, introducing the background of the artwork and the history of the Silk Road. To simulate the scene in the screen, we have created a physical miniature map where users can view the

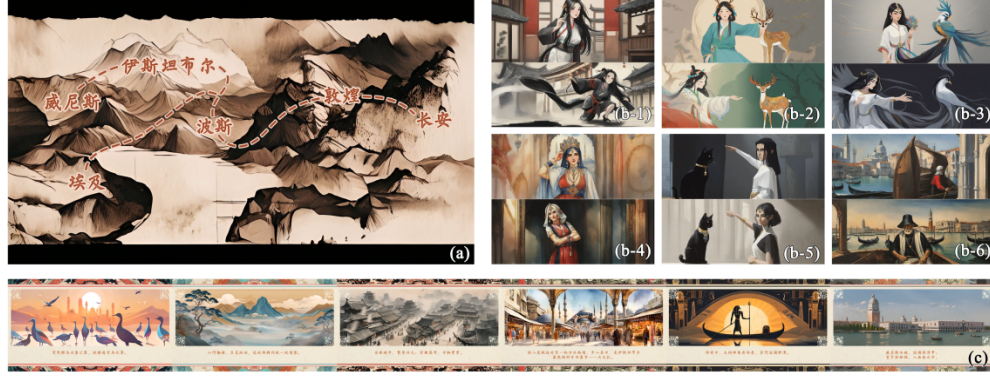


Figure 2: Silk Road Theme and Cultural Connotation: (a) Map Selection Interface (b) Stylized Images Generated in Real Time (c) Screenshot of Picture Book Content.

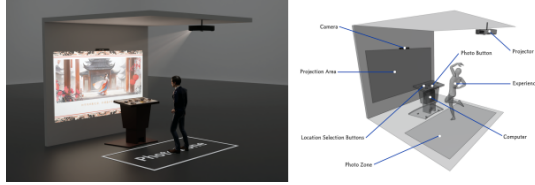


Figure 3: Installation layout

routes and select the locations they want to experience through buttons. In the interactive session, the player participates in the scenario through body gestures, manipulating the movements of the characters in the screen, thus influencing the direction of the whole story.

In terms of interactive design, in order to showcase different artistic styles and maintain a high level of generated image quality, we have imposed some limitations on user exploration, constraining the number of generated images and the variety of styles. However, we compensate for this limitation by integrating depth of interaction with the story plot. We encourage players to interact with the characters in the story. For instance, in the Dunhuang chapter, participants can touch, pet, or greet the Nine-Colored Deer, the protagonist of the graphic novel. In the Chang'an section, players are guided to take photos with traditional Chinese-style architecture as tourists. The images generated based on the players' postures are also incorporated as part of the picture book.

3.3 Installation Setup and Interaction Mode

The Silk Road Journey interactive setup consists of a projector with a projection screen (or high-definition LED screen), a 4K high-definition camera, a physical model of the Silk Road terrain, location selection and photo-taking buttons, speakers, and signs for the interactive area.

The interaction begins with the terrain rising, revealing a map that includes the Silk Road route and the locations that can be interacted with. Viewers enter the interactive area and choose the story segment they want to experience. Subsequently, the screen displays the visual content. When entering specific interactive

segments, the screen is replaced with real-time images captured by the camera, indicating the participant to play the role of a character in the story or interact with the story scene. After deciding on appropriate actions, the participant can press the photo button to freeze the camera image, which serves as input to render an stylized image. At the end of the experience, viewers can download the generated image by scanning the QR code on the screen.

The installation supports both individual viewer experiences and allows 2-4 members to watch the visuals simultaneously and interact. The duration of experiencing a single story segment, including location selection, narration, and interactive photo-taking, is approximately between 30 to 60 seconds.

4 TECHNICAL REALIZATION

4.1 Pose Control and Condition Control

The Silk Road journey utilizes Python scripts to collect image data captured by the computer's webcam and utilizes the ControlNet preprocessor of Stable Diffusion to generate facial expressions and skeletal data. Parameters used to control the images include limb and hand control points, and normal data obtained using the Normal preprocessor to reflect positional data in space. The story elements, which are used to interact with the audience, are fixed by another set of conditional control systems and remain blank in the rest of the frame.

Through testing, it has been found that when capturing backgrounds with complex scenes, the use of normal data may have potential negative effects on the generated images. By adjusting its weight to 0-0.3 and adjusting the weight of the OpenPose model to 1-1.2, stable rendering control effects can be achieved.

4.2 Image Generation and Stylization

In the initial demo testing, we found that using the traditional workflow resulted in slow generation speed and poor image quality. Therefore, we build the generation workflow by connecting nodes in ComfyUI. With an RTX 4090 graphics card, generating a 1920x1080 pixel image with 30 steps of sampling takes approximately 6-8 seconds. By introducing the LCM fine-tuning model, we were able to reduce the number of sampling steps and shorten the AI

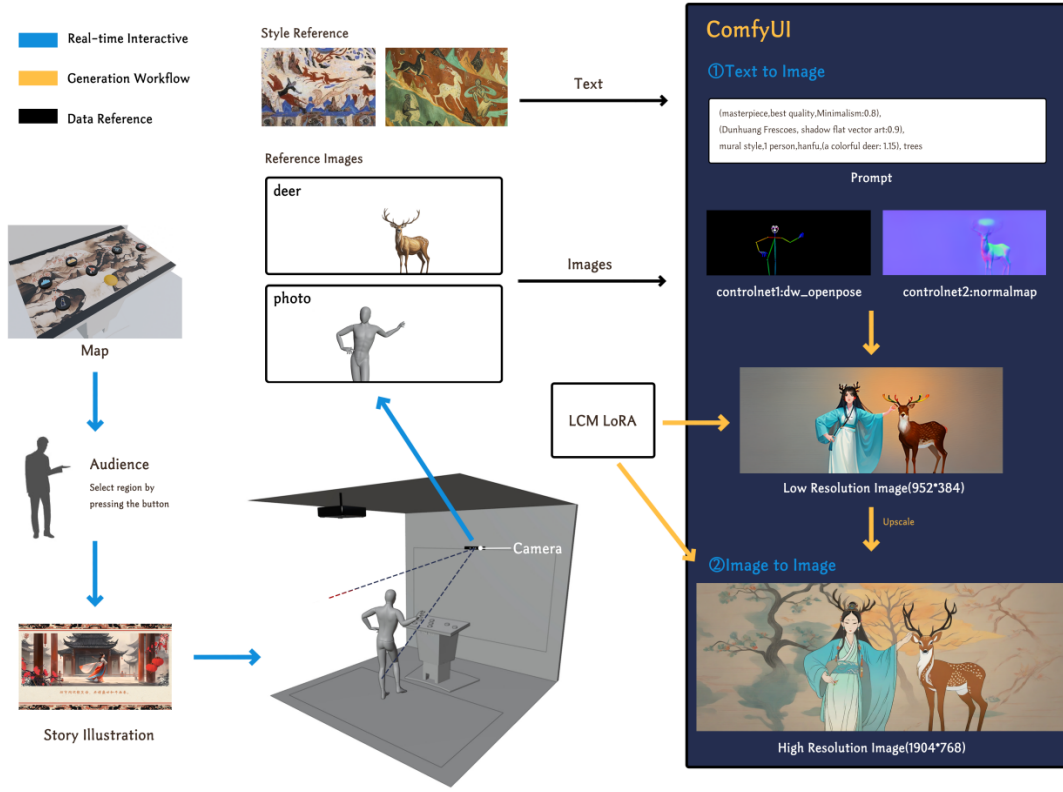


Figure 4: System Illustration. The system consists of two data flows: the real-time interactive (labeled in blue) and the ComfyUI generation workflow (labeled in yellow).

generation time to around 3 seconds, without compromising the quality of the generated images.

In generation tests, we found that the SDXL base model produces higher quality results, but limb alignment may be in error; whereas the ordinary Checkpoint model applies conditional control well. Therefore, to improve the quality of the generated images, the interactive installation combines these two approaches: the captured images are first transferred to the ControlNet OpenPose preprocessor, and the ordinary Checkpoint base model is used to generate low-resolution images. These images go through the VAE processor for decoding and re-encoding and then pass through the amplifier in the latent space vector, prompting the generation of high-resolution images using the SDXL. The redrawing ratio between the two generations is about 0.7-0.85, ensuring that the character pose remains unchanged and improving the rendering quality. The stylization of images for different regions is achieved by changing the base models as well as text and image prompts. For example, for the Chang'an region, we use an illustration-style base model and input keywords such as minimalism and ink painting to generate images with an ink wash painting style.

4.3 Real-time Data Transmission

Silk Road Journey implements ComfyUI real-time graph generation function by calling local API through Unity. Firstly, the pre-built workflow in ComfyUI's developer mode is saved as a JSON file. Then, the interfaces are used to communicate with Unity. The artwork utilizes three types of interfaces: POST /prompt is used to issue drawing tasks and returns the task ID information. GET /history can retrieve corresponding historical data based on the task ID. GET /view is used to obtain the generated images.

Depending on the chosen scene, different styles of JSON workflows are called when invoking the API. The real-time captured photos are used to replace the reference image in the ControlNet node, which is then inputted into ComfyUI to control the generation of images.

4.4 Interface Framework Construction

The interface framework of the interactive project is built based on Unity URP rendering pipeline, including three types of interfaces: map, generated storybook content display, and interactive photo-taking. Unity's UGUI game objects are used for layout and display of interface elements. Scripts are written in C# to implement interface navigation logic, control the movement and life cycle of game

objects, call external cameras, and transmit data to ComfyUI, among other functionalities.

5 CONCLUSION

This paper introduces the Silk Road Journey AI Interactive Installation, aiming to address the limited public awareness of the cultural heritage of the Silk Road, enhance cultural identity, and promote the publicity and preservation of transnational cultural heritage. By integrating AIGC technology, new media art and the protection of world cultural heritage, we provide new communication modes and experiential possibilities for the digital protection of cultural heritage. By exploring traditional culture and legendary stories, we enhance the immersion and engagement of the interaction. At the same time, we employ various strategies to improve the efficiency and quality of generating images based on posture control. Our preliminary research in the exhibition shows that the space not only engages audiences from different cultures, but also triggers their curiosity about the Silk Road culture. In the future, we plan to migrate the artwork from the local environment to online platforms, allowing audiences from around the world to experience the charm of the intersection of Silk Road culture and AIGC.

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